Agnes AI Track

AI for Science and Physical AI

WHAT YOU ARE BUILDING

Build a functional product that helps a defined user complete a scientific, engineering or physical-world task and demonstrate its usefulness with evidence.

How to use this brief

Choose one focused problem within this track. You are free to select the implementation, supporting tools and interface. A narrow workflow with a reproducible result is sufficient. The product directions below are starting points, not fixed problem statements. You may also propose another relevant scientific, engineering or physical-world problem with a clear user and evaluation method.

1. Track Overview

Build a working product that uses Agnes AI to help a defined user solve a scientific, engineering or physical-world problem. Choose a focused task, connect AI outputs to tools, data or simulation, and show measurable value against a baseline.

Track	AI for Science and Physical AI, including scientific research, engineering optimization and disaster-response support.
Challenge type	Product Innovation / New Product Development. Enhancements to an existing product are welcome when the team identifies its new contribution.
Technology	Agnes AI offers public text, image and video APIs that teams can use within scientific and engineering workflows.

2. Context and Problem to Solve

Scientific and engineering work often involves moving between documents, data, code and specialist tools. This track asks teams to connect an AI capability to a concrete user task and validate the output through domain tools, data or simulation. The aim is a useful prototype that can support a later discussion with researchers or potential users.

How can AI help a researcher, engineer or response planner complete a specific task more effectively, while making its outputs possible to check? Choose one user, one workflow bottleneck and one measurable outcome.

Consider the following when defining your problem:

• **Problem and users:** Researchers, engineers and planners can spend substantial time translating requirements into experiments, code, analyses or plans. Choose a concrete bottleneck rather than treating an entire scientific field as the problem.

• **Workflow stage:** Focus on a defined stage of a real or realistic workflow, such as validating a small design, tuning a compute kernel, reviewing protein research or planning a simulated rescue response.

• **Current limitations:** Manual handoffs and isolated AI responses can leave users with outputs they still need to verify or translate into a usable result. Identify the specific limitation in your baseline.

• **Evidence:** Substantiate the problem with cited domain material, a representative task or user feedback. Compare your prototype and baseline using the same inputs and evaluation conditions.

3. Suggested Product Directions

Choose one direction and narrow it to a specific user, task and measurable outcome. The examples below describe possible starting points; teams define their own solution.

01 Chip Design & Verification

Help users turn a small set of design requirements into a candidate implementation or testbench, then check it with permitted simulation or verification tools. A suitable outcome is fewer invalid designs or less time to reach a passing test case.

02 Operator Optimization

Help developers improve a selected numerical or AI operator. Measure runtime against an initial implementation on the same hardware and inputs, while preserving numerical correctness.

03 Protein Synthesis Research Support

Help researchers organize published evidence or permitted sequence data around a narrowly defined protein research question. Provide traceable sources and checks for accuracy. Wet-lab synthesis and biological validation are outside the required prototype.

04 Disaster Response & Rescue Planning

Help planners turn public or synthetic incident information into a reviewed situation summary, resource plan or simulated search route. Evaluate under stated scenario constraints and retain human review.

4. Users and Product Opportunity

Potential users	Researchers, engineering teams, scientific developers and disaster-response planners. Select a specific user and workflow.
Other stakeholders	Domain experts, research supervisors, infrastructure operators and organizations that could test or adopt the product.
Current workflow	The user gathers data and requirements, chooses a method, runs tools or simulations, checks results and decides what to do next. Identify the step your product will improve.
Common pain points	Repeated manual work, fragmented tools, hard-to-check AI outputs and limited time or computing resources for testing alternatives.
User outcome	Help the chosen user move from a task or question to a checked output that can be used in the next step of their workflow.
Meaningful improvement	Show improvement in correctness, speed, effort, cost or planning quality relative to a stated baseline. Choose the metric that fits your task.
Distinct value	Combine AI capability with a usable workflow and credible validation. Explain the contribution beyond an isolated model response or existing tool.

5. Expected Product Capabilities

Must-have	A working user workflow; meaningful use of at least one Agnes AI model in the product; traceable inputs and outputs; and a repeatable way to evaluate the result. Coding assistance alone does not demonstrate Agnes-powered product functionality.
Nice-to-have	Integration with domain tools or simulation, source citations, experiment history, uncertainty reporting or improvement across repeated attempts. Additional modalities are optional.
User interaction	A user supplies a task, data or scenario, reviews the proposed output and supporting evidence, and can revise, approve or reject the result. Teams choose their own interface and architecture.
What to demonstrate	A GitHub repository accessible for review, a README showing where Agnes is used and which models power each feature, and a short working demo. Include setup instructions, a reproducible test case, baseline results, limitations and known failure cases.

6. Technical and Operational Constraints

API limits and reliability	Free/default-account requests per minute (RPM) are dynamic and depend on model, resolution and platform capacity; use the current console limits [1]. Credits do not remove rate limits or guarantee capacity. Handle HTTP 429 and temporary service failures. Video generation can return HTTP 503 or a queue-full response during capacity pressure; allow bounded retries with backoff and a clear failure state [2].
Performance and accuracy	Define task-specific targets and compare against a stated baseline using a documented test set. Report correctness or quality alongside time, API usage and cost where relevant. Video is optional; prepare a clearly labelled recording of an actual run if live generation is delayed.
Security and data	Keep API keys server-side or in protected local configuration; exclude secrets from repositories and demos. Use public, synthetic or explicitly permitted data. Identify dataset licenses and obtain permission before using personal or confidential material.
Hardware and deployment	API calls require internet access. Teams provide their own computing environment and domain-specific tools. No dedicated GPU, robot, laboratory, chip fabrication facility or private research API is included. Simulation and small-scale tests are acceptable.
Safety	Use simulation, public data and expert review appropriate to the task. Scientific outputs require independent validation. Disaster-response and robotics demonstrations must remain supervised prototypes, with human approval before consequential actions. Wet-lab work, real emergency dispatch and autonomous operation in hazardous settings are not required.

7. Agnes AI Resources

Current published free offers cover Agnes 2.5 Flash and 3.0 Flash text models, Flash image models and Video 2.5 Flash. Check pricing and account permissions before use [3]; rate limits and capacity still apply.

Rounds 1 and 2	Use the existing free API offering under its current availability and limits.
Finalist API balance	The 12 finalist teams receive US$10 per registered member, credited individually. A three-member team therefore has US$30 in aggregate balances; a four-member team has US$40. Each balance remains on its own account.
Account setup	Each finalist member must create an account at platform.agnes-ai.com and provide their registered email for credit allocation. Passwords and API keys are not required by the organiser.
Credits	Sponsored balances do not expire and remain available after the event until used. Credit activation timing is coordinated by the organiser. Further funding is not committed.
Data	Teams may use public, synthetic or properly licensed datasets that fit their chosen problem. Agnes AI has not committed a proprietary scientific dataset; document data provenance and permitted use.

Development environment	Teams may build local test environments or simulations. A dedicated event sandbox, reserved inference capacity and specialist hardware are not currently committed.
Documentation and tools	Public API documentation, model guides, pricing and error-handling references [1–3]. Agnes Harness is available as an optional public development resource [4]; using it does not imply access to private research systems.

8. How to Evaluate Your Product

Demonstrate success on a bounded task with a stated baseline, reproducible inputs and relevant measurements. The final judging rules are to be confirmed by the organiser.

Functional result	Demonstrate an end-to-end workflow on a reproducible case, including actual Agnes model use and handling of invalid inputs or API failure.
User value	Explain who benefits and compare effort, usability or decision quality with the current workflow. Include user or expert feedback where available; identify assumptions where it is unavailable.
Technical evidence	Report a relevant baseline and metric, such as test pass rate for generated code, operator runtime at equal numerical correctness, source accuracy for research assistance, or planning results under a simulated scenario. Include limitations and failures, not only the best run.
Cost and feasibility	Report approximate cost and time per completed task and explain what would be needed for a small pilot.
Responsible design	Show data provenance, protected credentials, clear uncertainty and appropriate human review. Distinguish generated suggestions from validated results. Report retries and failures for workflows that depend on external APIs.

9. Scope and Boundaries

In scope	A focused scientific, engineering or physical-world workflow with meaningful Agnes model use, a working prototype and measurable evaluation. Third-party tools, open-source software and simulations may be used as permitted by their licenses and event rules.
Out of scope	A generic chat wrapper without a defined domain task; a concept deck without working functionality; unsupported claims of scientific discovery, production readiness or autonomous rescue. Full-scale fabrication, wet-lab validation and training a foundation model are not required.
Acceptable simplifications	Small datasets, toy circuits, CPU-scale benchmarks, synthetic incidents and simulated environments are acceptable. State assumptions, label prerecorded demonstrations, and disclose how a simplified test differs from real deployment.
Dependencies	Registered API accounts for finalist credit allocation; internet access; permitted datasets and external tools; and the organiser's submission rules.

10. Business and Adoption Context

Explain who could use your product after the event and what evidence would justify a small follow-up test.

Alternatives	Compare with the chosen manual workflow, existing domain software or a simple AI baseline. Name an appropriate alternative and compare the same task under similar conditions.
Potential value	Reduced task time or repetitive work, better engineering performance at equal correctness, improved evidence checking, or more useful planning outputs. Quantify the benefit your prototype actually demonstrates.
Adoption pathway	Validate the prototype with a relevant user or expert, document remaining gaps, and identify a small follow-up test. Further discussion with Agnes AI researchers or partners is possible by mutual interest and is not guaranteed.

REFERENCE DOCUMENTATION

[1] Access plans and current RPM rules
[2] API integration and error handling
[3] Current model availability and pricing
[4] Agnes Harness public repository
Agnes AI API platform